

PVsyst - Simulation report

Grid-Connected System

Project: Yunus Textile 9MW

Variant: New simulation variant

No 3D scene defined, no shadings

System power: 9000 kWp

Karachi - Pakistan



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Project summary

Geographical Site

Karachi
Pakistan

Situation

Latitude 24.83 °N
Longitude 67.17 °E
Altitude 33 m
Time zone UTC+5

Project settings

Albedo 0.20

Weather data

Karachi
Meteonorm 8.1 (1996-2015), Sat=100% - Synthetic

System summary

Grid-Connected System

No 3D scene defined, no shadings

PV Field Orientation

Fixed plane
Tilt/Azimuth 15 / 0 °

Near Shadings

No Shadings

User's needs

Fixed constant load
6500 kW
Global
56.94 GWh/Year

System information

PV Array

Nb. of modules 15385 units
Pnom total 9000 kWp

Inverters

Nb. of units 72 units
Pnom total 9000 kWac
Grid power limit 1000 kWac
Grid lim. Pnom ratio 9.000

Results summary

Produced Energy	15083140 kWh/year	Specific production	1676 kWh/kWp/year	Perf. Ratio PR	82.51 %
Used Energy	56939994 kWh/year			Solar Fraction SF	26.41 %

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General parameters

Grid-Connected System

No 3D scene defined, no shadings

PV Field Orientation

Orientation

Fixed plane

Tilt/Azimuth 15 / 0 °

Sheds configuration

No 3D scene defined

Models used

Transposition Perez
Diffuse Perez, Meteonorm
Circumsolar separate

Horizon

Free Horizon

Near Shadings

No Shadings

User's needs

Fixed constant load
6500 kW
Global
56.94 GWh/Year

Grid power limitation

Active power 1000 kWac
Pnom ratio 9.000

PV Array Characteristics

PV module

Manufacturer Jinkosolar
Model JKM-585N-72HL4-BDV
(Original PVsyst database)

Unit Nom. Power 585 Wp
Number of PV modules 15385 units
Nominal (STC) 9000 kWp
Modules 905 string x 17 In series

At operating cond. (50°C)

Pmpp 8347 kWp
U mpp 671 V
I mpp 12434 A

Total PV power

Nominal (STC) 9000 kWp
Total 15385 modules
Module area 39743 m²

Inverter

Manufacturer Huawei Technologies
Model SUN2000-125KTL-M0
(Original PVsyst database)

Unit Nom. Power 125 kWac
Number of inverters 72 units
Total power 9000 kWac
Operating voltage 200-1000 V
Max. power (=>30°C) 138 kWac
Pnom ratio (DC:AC) 1.00
Power sharing within this inverter

Total inverter power

Total power 9000 kWac
Max. power 9900 kWac
Number of inverters 72 units
Pnom ratio 1.00

Array losses

Array Soiling Losses

Loss Fraction 3.0 %

Thermal Loss factor

Module temperature according to irradiance
Uc (const) 20.0 W/m²K
Uv (wind) 0.0 W/m²K/m/s

DC wiring losses

Global array res. 0.88 mΩ
Loss Fraction 1.5 % at STC

Module Quality Loss

Loss Fraction -0.8 %

Module mismatch losses

Loss Fraction 2.0 % at MPP

IAM loss factor

Incidence effect (IAM): Fresnel, AR coating, n(glass)=1.526, n(AR)=1.290

0°	30°	50°	60°	70°	75°	80°	85°	90°
1.000	0.999	0.987	0.962	0.892	0.816	0.681	0.440	0.000



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AC wiring losses

Inv. output line up to injection point

Inverter voltage	500 Vac tri
Loss Fraction	1.50 % at STC

Inverter: SUN2000-125KTL-M0

Wire section (72 Inv.)	Copper 72 x 3 x 50 mm ²
Average wires length	82 m



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Main results

System Production

Produced Energy	15083140 kWh/year	Specific production	1676 kWh/kWp/year
Used Energy	56939994 kWh/year	Perf. Ratio PR	82.51 %
		Solar Fraction SF	26.41 %

Economic evaluation

Investment

Global	3,150,078.75 USD
Specific	0.35 USD/Wp

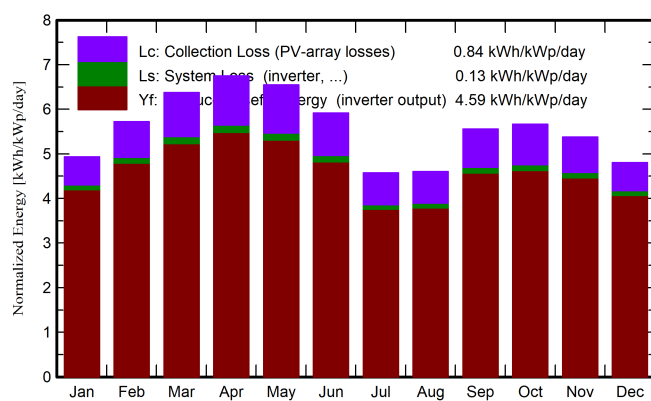
Yearly cost

Annuities	0.00 USD/yr
Run. costs	45,001.13 USD/yr
Payback period	1.3 years

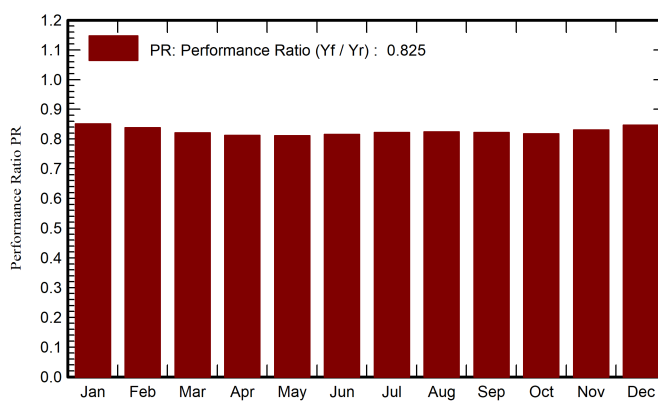
LCOE

Energy cost	0.01 USD/kWh
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Normalized productions (per installed kWp)



Performance Ratio PR



Balances and main results

	GlobHor	DiffHor	T_Amb	GlobInc	GlobEff	EArray	E_User	E_Solar	E_Grid	EFrGrid
	kWh/m²	kWh/m²	°C	kWh/m²	kWh/m²	kWh	kWh	kWh	kWh	kWh
January	125.0	46.2	18.60	153.1	145.4	1202703	4835999	1171427	265	3664573
February	137.7	50.1	21.54	160.2	152.5	1242369	4368000	1204986	3596	3163014
March	181.5	69.1	26.28	197.7	188.1	1503808	4835999	1445659	15090	3390340
April	197.0	78.9	29.18	202.4	192.8	1525370	4679999	1465112	15410	3214887
May	207.2	97.0	31.32	203.0	192.8	1525977	4835999	1479592	2515	3356408
June	183.9	103.2	31.59	177.5	168.5	1341155	4679999	1303354	404	3376645
July	146.0	97.8	30.70	141.9	134.2	1078355	4835999	1048155	1397	3787845
August	142.9	91.5	29.40	142.7	135.2	1088159	4835999	1056647	2094	3779353
September	158.7	82.5	29.14	166.8	158.4	1269760	4679999	1230553	4061	3449447
October	155.9	66.5	29.33	175.6	167.0	1328937	4835999	1290284	1454	3545715
November	133.0	44.2	24.68	161.3	153.5	1239481	4679999	1205226	459	3474774
December	119.4	41.2	20.10	149.0	141.7	1165710	4835999	1135414	-13	3700585
Year	1888.0	868.2	26.85	2031.2	1930.1	15511784	56939994	15036408	46732	41903585

Legends

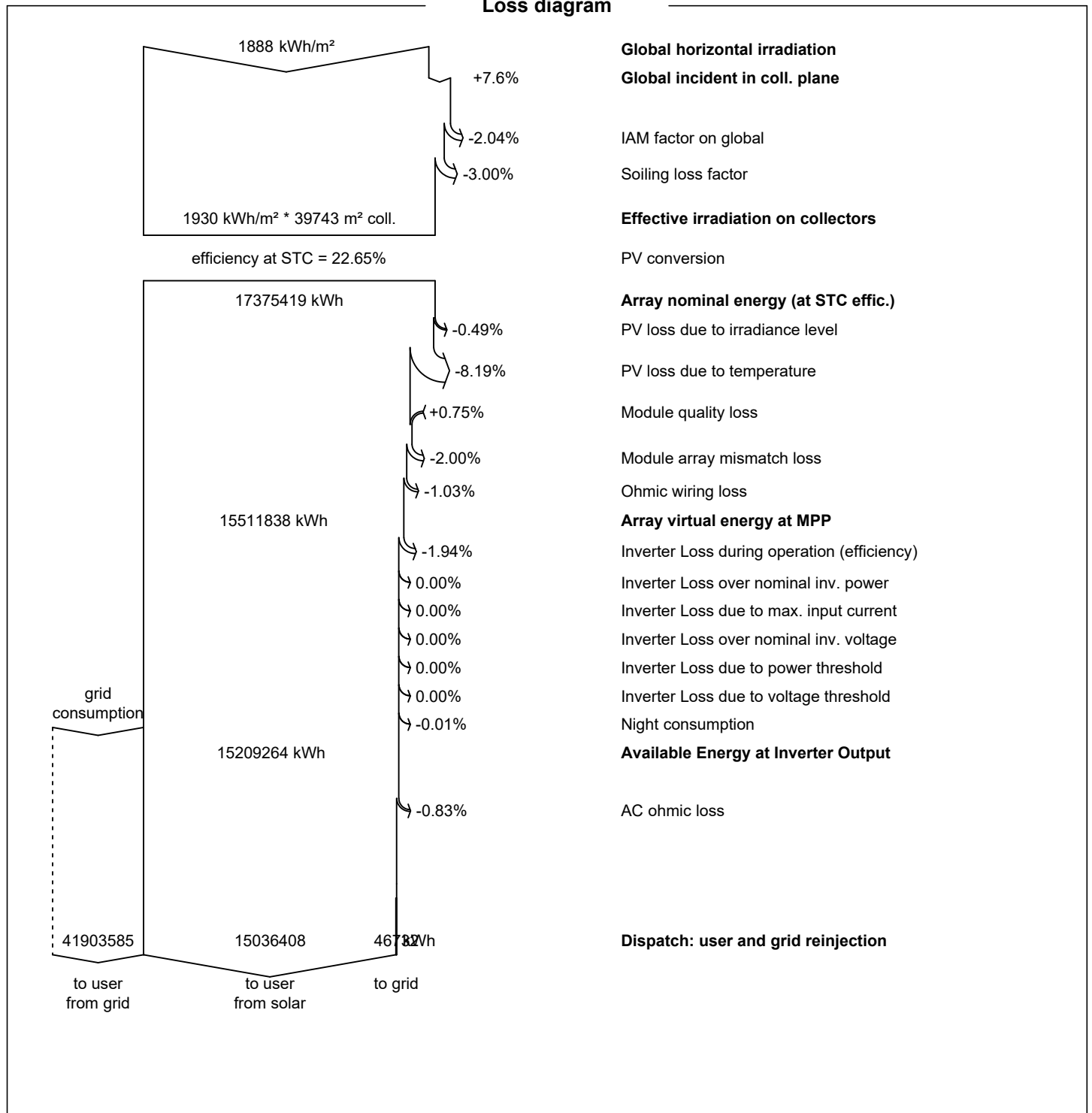
GlobHor	Global horizontal irradiation	EArray	Effective energy at the output of the array
DiffHor	Horizontal diffuse irradiation	E_User	Energy supplied to the user
T_Amb	Ambient Temperature	E_Solar	Energy from the sun
GlobInc	Global incident in coll. plane	E_Grid	Energy injected into grid
GlobEff	Effective Global, corr. for IAM and shadings	EFrGrid	Energy from the grid



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Loss diagram



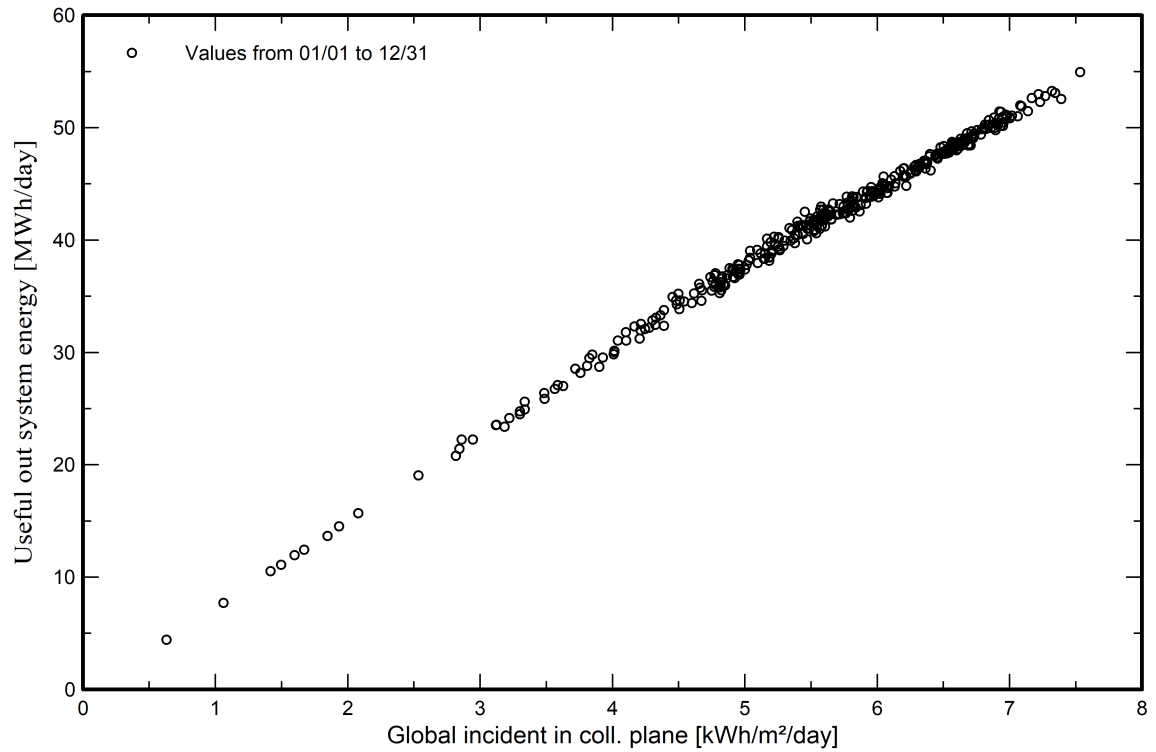


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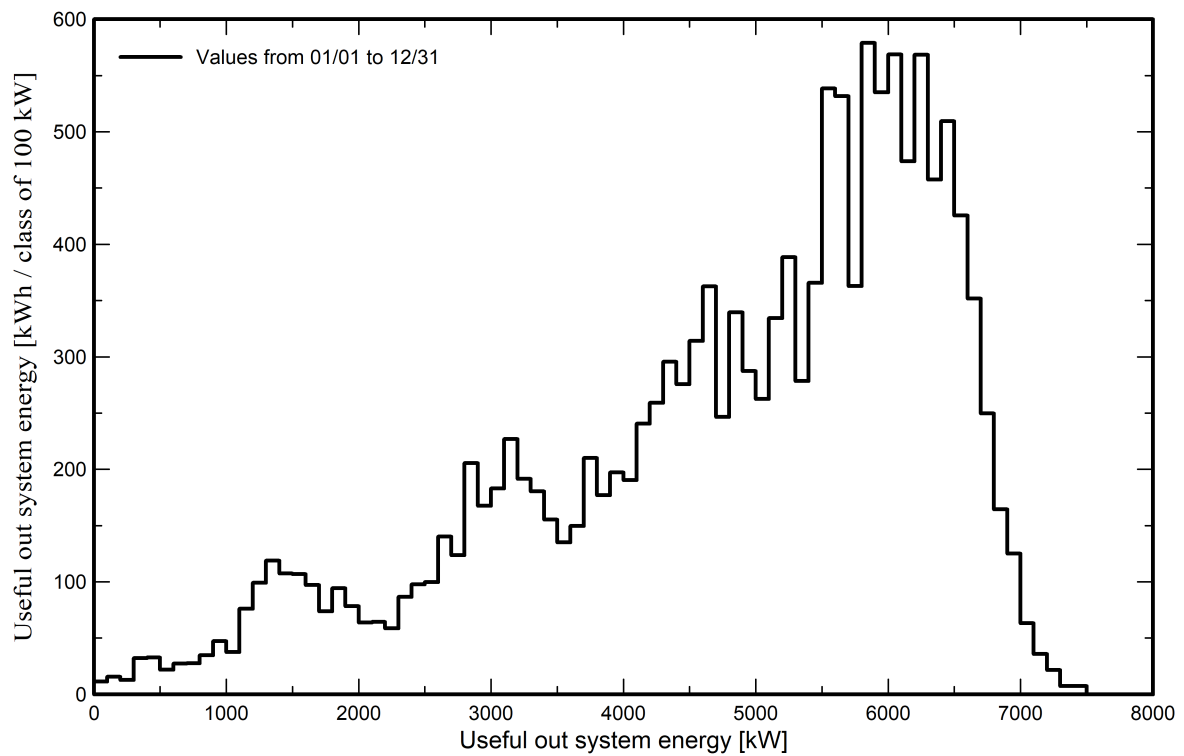
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Predef. graphs

Daily Input/Output diagram



System Output Power Distribution

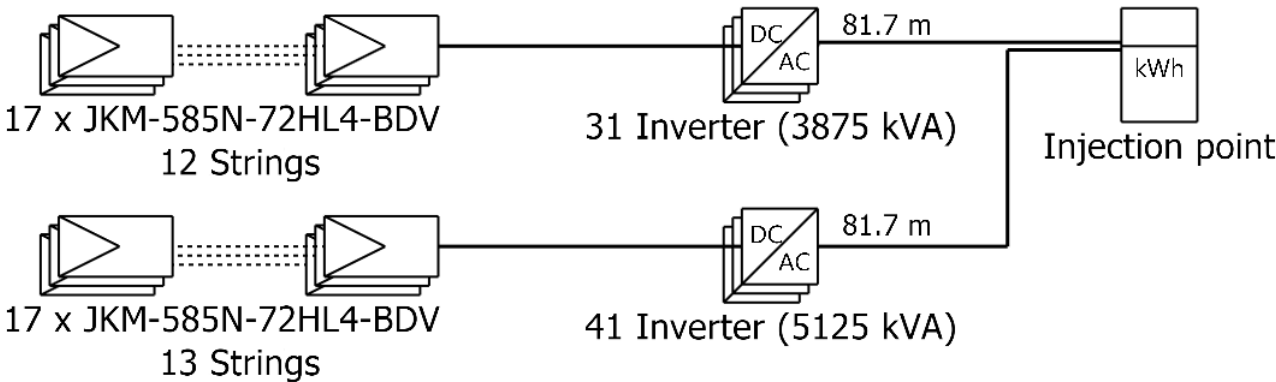




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Single-line diagram



PV module	JKM-585N-72HL4-BDV
Inverter	SUN2000-125KTL-M0
String	17 x JKM-585N-72HL4-BDV

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01/24/26



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Cost of the system

Installation costs

Item	Quantity units	Cost USD	Total USD
PV modules JKM-585N-72HL4-BDV	15385	204.75	3,150,078.75
		Total	3,150,078.75
		Depreciable asset	3,150,078.75

Operating costs

Item	Total USD/year
Maintenance	
Cleaning	45,001.13
Total (OPEX)	45,001.13

System summary

Total installation cost	3,150,078.75 USD
Operating costs	45,001.13 USD/year
Useful energy from solar	15036 MWh/year
Energy sold to the grid	46.7 MWh/year
Cost of produced energy (LCOE)	0.0134 USD/kWh



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Financial analysis

Simulation period

Project lifetime 20 years Start year 2027

Income variation over time

Inflation 0.00 %/year
Production variation (aging) 0.00 %/year
Discount rate 0.00 %/year

Income dependent expenses

Income tax rate 0.00 %/year
Other income tax 0.00 %/year
Dividends 0.00 %/year

Depreciable assets

Asset	Depreciation method	Depreciation period (years)	Salvage value (USD)	Depreciable (USD)
PV modules JKM-585N-72HL4-BDV	Straight-line	20	0.00	3,150,078.75
		Total	0.00	3,150,078.75

Financing

Own funds 3,150,078.00 USD

Electricity sale

Feed-in tariff 0.16000 USD/kWh
Duration of tariff warranty 20 years
Annual connection tax 0.00 USD/kWh
Annual tariff variation 0.0 %/year
Feed-in tariff decrease after warranty 0.00 %

Self-consumption

Consumption tariff 0.16000 USD/kWh
Tariff evolution 0.0 %/year

Return on investment

Payback period 1.3 years
Net present value (NPV) 44,219,565.60 USD
Internal rate of return (IRR) 75.19 %
Return on investment (ROI) 1403.8 %



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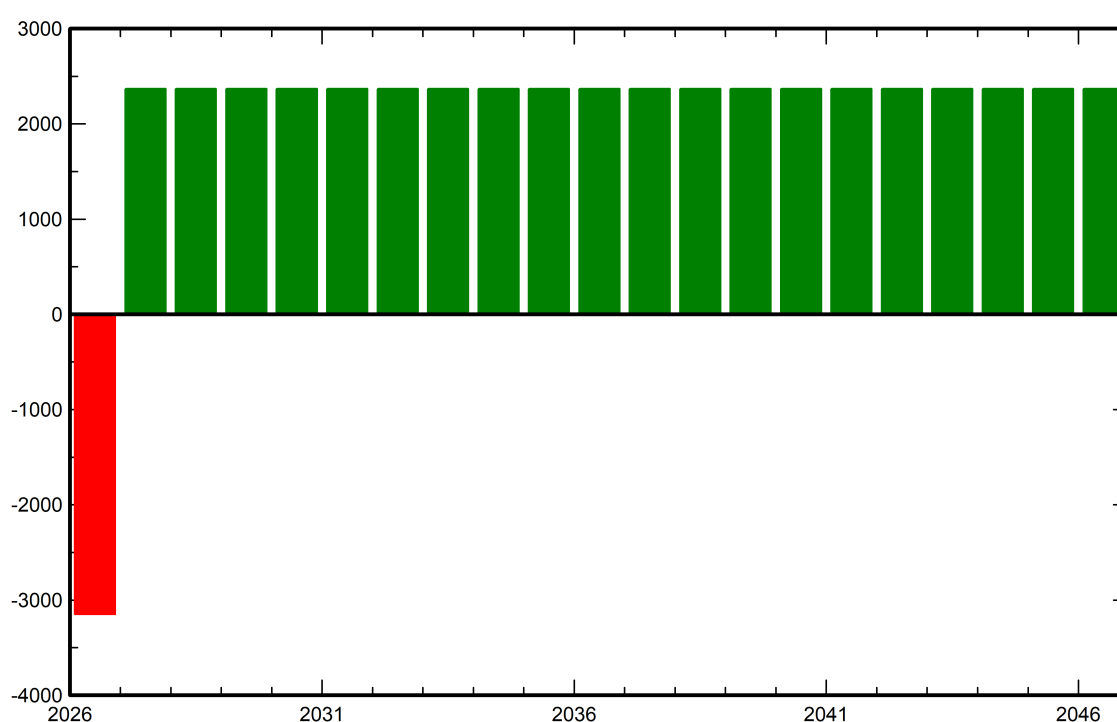
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Financial analysis

Detailed economic results (USD)

Year	Electricity sale	Own funds	Run. costs	Deprec. allow.	Taxable income	Taxes	After-tax profit	Self-cons. saving	Cumul. profit	% amorti.
0	0	3,150,078	0	0	0	0	0	0	-3,150,078	0.0%
1	7,658	0	45,001	157,504	0	0	-37,343	2,405,826	-781,596	75.2%
2	7,658	0	45,001	157,504	0	0	-37,343	2,405,826	1,586,886	150.4%
3	7,658	0	45,001	157,504	0	0	-37,343	2,405,826	3,955,369	225.6%
4	7,658	0	45,001	157,504	0	0	-37,343	2,405,826	6,323,851	300.8%
5	7,658	0	45,001	157,504	0	0	-37,343	2,405,826	8,692,333	375.9%
6	7,658	0	45,001	157,504	0	0	-37,343	2,405,826	11,060,815	451.1%
7	7,658	0	45,001	157,504	0	0	-37,343	2,405,826	13,429,297	526.3%
8	7,658	0	45,001	157,504	0	0	-37,343	2,405,826	15,797,779	601.5%
9	7,658	0	45,001	157,504	0	0	-37,343	2,405,826	18,166,262	676.7%
10	7,658	0	45,001	157,504	0	0	-37,343	2,405,826	20,534,744	751.9%
11	7,658	0	45,001	157,504	0	0	-37,343	2,405,826	22,903,226	827.1%
12	7,658	0	45,001	157,504	0	0	-37,343	2,405,826	25,271,708	902.3%
13	7,658	0	45,001	157,504	0	0	-37,343	2,405,826	27,640,190	977.4%
14	7,658	0	45,001	157,504	0	0	-37,343	2,405,826	30,008,673	1052.6%
15	7,658	0	45,001	157,504	0	0	-37,343	2,405,826	32,377,155	1127.8%
16	7,658	0	45,001	157,504	0	0	-37,343	2,405,826	34,745,637	1203.0%
17	7,658	0	45,001	157,504	0	0	-37,343	2,405,826	37,114,119	1278.2%
18	7,658	0	45,001	157,504	0	0	-37,343	2,405,826	39,482,601	1353.4%
19	7,658	0	45,001	157,504	0	0	-37,343	2,405,826	41,851,083	1428.6%
20	7,658	0	45,001	157,504	0	0	-37,343	2,405,826	44,219,566	1503.8%
Total	153,154	3,150,078	900,023	3,150,079	0	0	-746,868	48,116,512	44,219,566	1503.8%

Yearly net profit (kUSD)





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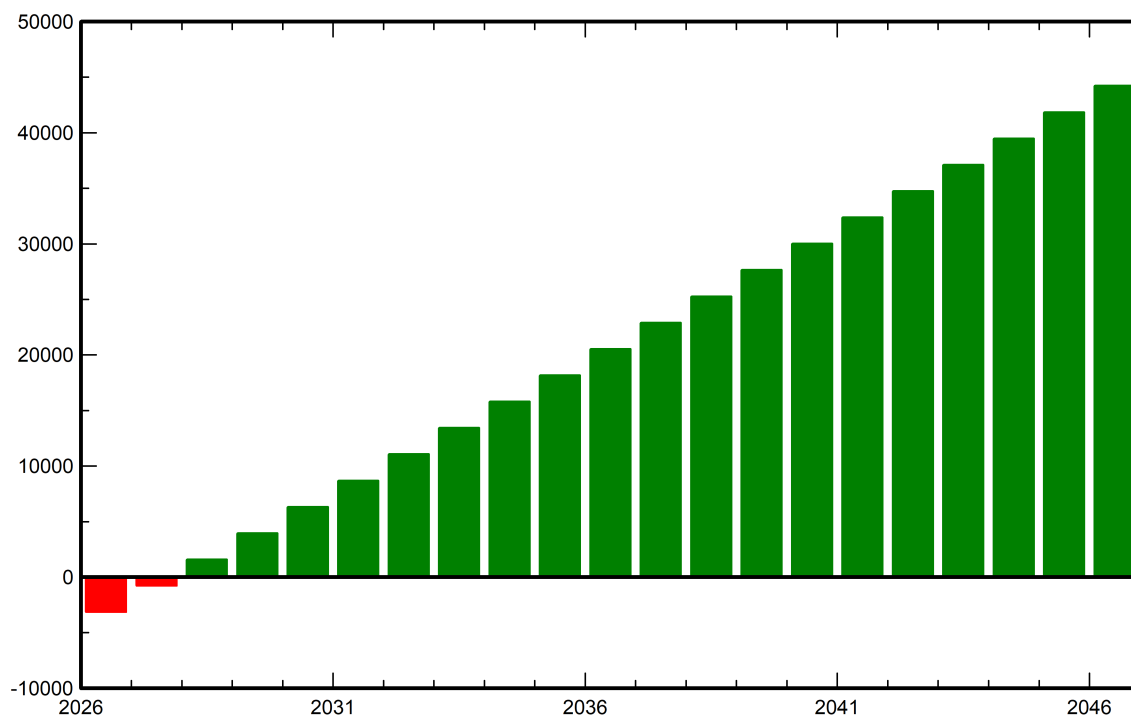
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Financial analysis

Cumulative cashflow (kUSD)





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CO₂ Emission Balance

Total: 158833.4 tCO₂

Generated emissions

Total: 15892.69 tCO₂

Source: Detailed calculation from table below

Replaced Emissions

Total: 201375.0 tCO₂

System production: 15084.27 MWh/yr

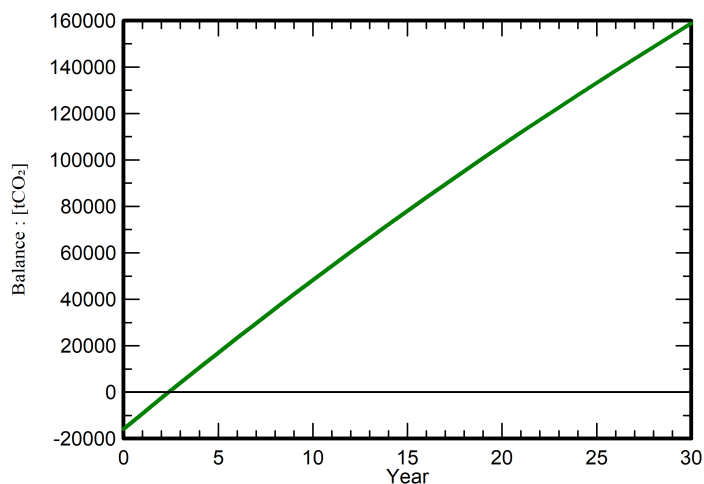
Grid Lifecycle Emissions: 445 gCO₂/kWh

Source: IEA List

Country: Pakistan

Lifetime: 30 years

Annual degradation: 1.0 %

Saved CO₂ Emission vs. Time**System Lifecycle Emissions Details**

Item	LCE	Quantity	Subtotal
			[kgCO ₂]
Modules	1713 kgCO ₂ /kWp	9000 kWp	15414865
Supports	2.97 kgCO ₂ /kg	153850 kg	456650
Inverters	294 kgCO ₂ /units	72.0 units	21172